Presentation script and speaking notes

Student dropout warnings

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Slides 1–12 are the main talk. Slide 13 contains sources and the AI-use acknowledgement. A possible split for four speakers is slides 1–3, 4–6, 7–9 and 10–12. Confirm the allocation and timing in rehearsal.

Slide 1 · Student dropout warnings

Who gets missed when we apply one dropout warning rule to every student? To answer that, we need to know what counts as dropout, what we know at enrolment, and how a risk score becomes a warning. We start with the outcome that we can observe.

Slide 2 · What counts as recorded dropout?

We used 4,424 records from one Portuguese institution. The outcome is recorded at the normal course endpoint, typically after three years. A change of course or institution can count as dropout. Graduate and Enrolled both mean no recorded dropout at that endpoint. With the outcome defined, what could we have known when students first enrolled?

Slide 3 · The prediction starts at enrolment

The prediction is made at enrolment, using age, admission grade and course. The recorded outcome is observed at the normal course endpoint. Later grades cannot be used to make an enrolment prediction, and financial fields with unclear timing were omitted. Now look at how the available inputs vary across outcome groups.

Slide 4 · What do we know at enrolment?

These plots use training records. Students later recorded as dropouts were typically older, but the distributions overlap, including for admission grade. The counts under each box show the number of training records. Because single inputs only tell part of the story, the next slide shows how we combine age, admission grade and course into one risk score.

Figure review: Quiz 7 recommended Data to Viz. We wanted a compact comparison of the centres and spread. Its boxplot page reminded us that equal-width boxes hide group sizes, so we added the training count below each outcome. We retained boxplots for this comparison; they do not reveal the full distribution. Each panel has its own labelled units. This is our design choice, not an additional course requirement.

Slide 5 · From enrolment to a warning

The model combines age, admission grade and course to estimate a probability of recorded dropout. Once a threshold is set, a score at or above it produces a warning. We fit on 2,652 training records, chose the threshold on 883 validation records and evaluated the frozen rule on 889 test records. Recorded gender is used in the group audit. So what score should trigger a warning?

Original R formula for Q&A: `dropout ~ age10 + admission10 + course .`

Slide 6 · What score triggers a warning?

On validation records, we chose the threshold with the highest specificity among thresholds that found at least 70 percent of recorded dropouts. At 0.31 it found 200 of 284. We then used that same threshold on the test data. The gold vertical line marks 0.31 and the dotted horizontal line marks the 70 percent sensitivity target. What changes when we lower the threshold from 0.50 to 0.31?

Figure review: We kept lines because the horizontal axis is a numerical threshold and the purpose is to compare how rates change. After checking Data to Viz's colour caveats, we gave all three curves different line patterns as well as colours. The curves remain descriptive validation rates, not confidence bands; the full 0-100% rate scale is retained.

Slide 7 · A lower threshold changes who gets warned

These are comparisons on the same 889 held-out records, using the same model. With 0.31 instead of 0.50, the rule flagged 107 additional records with a recorded dropout and 177 additional records without one. Both changes come from the threshold, because the risk scores stayed fixed. The next slide reports the overall performance of the rule chosen on validation data.

Slide 8 · How many dropouts did we miss?

On 889 test records, the 0.31 rule found 187 of 285 recorded dropouts and produced 222 false warnings. The chosen rule still missed 98 recorded dropouts. How were those 98 misses distributed across recorded gender groups?

Slide 9 · The missed-dropout rate differs by recorded gender

These groups account for all 98 missed records. To compare the rates, divide by the number of recorded dropouts within each gender group. We estimate a 20.3 percentage point gap, with a 95 percent bootstrap interval conditional on the fixed model and threshold. This describes the held-out data. The next slide checks whether the same pattern appears for false warnings.

Slide 10 · Who was more often warned without a recorded dropout?

The previous slide showed that the rule missed a larger share of recorded dropouts in the female group. Now look at a different denominator: students with no recorded dropout. Of these, 138 of 431 in the recorded female group and 84 of 173 in the recorded male group were flagged. The same rule can miss students who later drop out and flag students who do not. Do we see a similar trade-off when we group students by whether they left home to study?

Slide 11 · Do the error patterns extend beyond gender?

After comparing recorded gender, we applied the same two thresholds to a second grouping, the dataset's Displaced field. At 0.31, the group marked displaced had more missed dropouts proportionally, but fewer false warnings proportionally. The same directions appeared at 0.50. Both groups missed more dropouts at the higher threshold. These are descriptive differences in model errors. Before anyone acts on a warning, the student's circumstances still need checking.

Slide 12 · Review the warning before offering support

The two error patterns explain why this needs human review. A flagged student may benefit from a check-in even if they never receive a dropout label. An unflagged student should still be able to access support. Staff would confirm the records, ask about the student's needs and

offer support where appropriate. We would also check capacity and validate on later cohorts. This study measured prediction errors. It did not measure whether outreach prevents dropout.

Slide 13 · Sources and assistance

See the references and disclosure on the final slide and in the report draft.